

Mental Sketching Model (MSM)

(Project) Assistive Tools, AI, Design, Sketching, Mental Models

A Project Proposal

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Abstract

New designers adapt to use new tools and technologies, including artificial intelligence (AI). Use of AI currently requires designers to navigate a variety of platforms, applications or interfaces to produce their desired outcome. In the realm of sketching, AI-empowered tools like [18] have been developed to take drawings/sketches from novice designers and then produce the same work, albeit in higher fidelity with little to no additional effort on the part of the designer. While such tools assist designers in sketching directly, they also impact the final design output and can potentially insert hallucinations or misinterpretations into the final outcome of the design process. The final outcome should be more authentic and true to the intentions of the designer's Mental Model. This project proposes an alternative approach, where designers can intentionally limit how AI-influences the designers' working file, by creating a generative AI reference layer for designers to model from. In this Mental Sketching Model approach, the system takes user's working sketch and textual prompts to generate an exemplar sketch on a separate layer—as a form of visual feedback or reference to support their own design intention and iteration. The aim of this work is to support the development of essential design skills, shifting the focus from using generative AI as a mere output to learning from it through scaffolding practices and improving cognitive design workflows.

Introduction

Designers and non-designers alike often rely on sketches to externalize ideas, develop product concepts, and communicate their mental models to others. Sketching is not only a way to present a concept, but also a cognitive process through which people explore, refine, and construct their understanding of a design [33]. As multimodal systems become more capable of supporting creative work [18], there is an opportunity to use these systems not to replace sketching, but to augment it.

Sketching is a foundational practice for externalizing and refining mental models throughout the design process; however, many designers, particularly students and novices, struggle to iteratively translate internal intent into coherent visual representations [26]. While prior work has explored sketching as a learning practice [6], existing support mechanisms often rely on static instruction, skill-based training, or prescriptive tutorials, offering limited assistance during the moment-to-moment act of ideation and iteration [32].

Recent advances in generative AI image technologies introduce new opportunities for supporting sketch-based design [19, 8]. However, current implementations frequently prioritize automation, fidelity, or output generation, raising concerns around authorship, overreliance, and the substitution of human intent [3, 18]. Most related to this project is ShadowDraw, which dynamically updates a shadow image underlay of the user's strokes for proportional guidance, but it does not utilize generative AI for conceptually driven designs and relies on pre-existing images, which may limit new designs [17]. As a result, there exists a gap in

understanding how generative AI can be integrated into the sketching process as an assistive reference system that supports designers' iterative communication of mental models without directly influencing or altering final design outcomes.

This design research project addresses that gap through the development of an AI-based assistive technology titled *Mental Sketching Model (MSM)*. The project is motivated by the need to support human sketching capabilities, specifically for aiding designers in conceptualizing and expressing their mental models for product designs [33]. Rather than positioning generative AI as a design generator or editor, this work constrains its role to that of a reference-based assistive system. The focus is on enhancing the individual's capacity to more accurately sketch internal mental representations while preserving full creative agency and decision-making authority. Because mental models are internal representations, it can be difficult for novices to convey their concepts on a canvas due to their lack of skill [14, 19]. This project therefore proposes an abstracted "sketch scaffold" or reference layer that aids in a non-intrusive manner through scaffolding and discovery, rather than through instructional guides and tutorials that are more structured and generic [18]. In doing so, it also foregrounds ethical decision-making processes by preserving human intent and authorship throughout the design workflow [32].

The project is presented as a system that assists users in sketching interdependently through multimodal generative AI technologies. By incorporating both the designer's textual input and the current state of their sketch, this implementation of AI sketching tools is designed to support human-authored product designs without AI-based systems taking over human intent and authorship. This approach aligns with an *AI-in-the-Loop model* rather than a *Human-in-the-Loop model* [20].

MSM provides generative AI images as a reference underlay tailored to help individuals who are still developing their drawing skills by assisting them with translating their internal mental models into visually depicted concept models [19]. Because sketching with a stylus unfolds iteratively, stroke by stroke, the user engages with a continuously changing visual stimulus during the act of drawing [32]. This makes sketching especially well suited to a workflow in which generative AI functions as a dynamic scaffold rather than a direct producer of the final result.

Methodologically, this project follows a research through design approach [2] that combines landscape analysis and systematic literature review, as are common within Industrial Design practice and research (e.g., [13]) with direct observation of sketching practices, and iterative prototyping to understand current workflows and gaps in assistive design tools that utilize AI technologies.

Presented in Chapter 2, the literature review supported the development of a conceptual framework centered on three primary themes: 1) sketching and learning, which examines sketching as a cognitive and pedagogical

practice for externalizing and refining mental models [12, 19]; 2) assistive technologies and cognitive impact, which explores how assistive systems influence cognition, agency, and creative ownership/authorship [8]; and 3) AI usage as a pedagogical tool, which positions AI as an active partner in design education [19]. Together, these themes established an analogical connection between sketching, its effects on users' mental models, and the foundational requirements for the system.

Building from this framework, and informed by Natarajan et al. [20], the iterative design process translated the literature review's key insights into successive design decisions that shaped the system's development.

A central question emerged through this process: *How can AI technology be utilized and interacted with by the user, in ways that preserve their intent and ability, while gaining assistance in the sketch process?*

In turn, the primary focus for this project was to design an interactive and collaborative workspace and interface design for the designer to engage with the AI. A key consideration was to combat design fixation that occurs when a user is presented with high-fidelity generated imagery too early in the ideation process [17]. This project applies a novel usage of an AI implemented workflow that instead relies on lower fidelity renderings so as to not lead the designer into any design fixation. This encourages the use of this application as a tool to assist the designer with their conceptualization without limiting their conceptualization.

To overcome design fixation, MSM utilizes a "Sketching Scaffold" supportive tool that mitigates fixation by abstracting high-fidelity generative outputs into lower-resolution, sketch-like translucent underlays based on the user's text and image inputs [17, 18]. This layered approach adapts the classical technique of transparent tracing underlays into a multimodal digital workflow, enabling participants to make frequent small adjustments rather than relying on the AI to simply output text-to-image prompts. The AI layer remains separate and cannot be exported into the final outcome. The designer must actively trace, modify, align, or intentionally ignore the AI's interpretations as they sketch to reach their intended design goals. In this way, the system preserves subjective agency while enabling a reflective conversation with the reference material (that being the AI generated layer) [18,10].

This project therefore functions not only as a technical output, but as a scaffold for ethical Human-AI collaboration in design practice, with outcomes expressed through both research insights and the implementation of an interactive functional prototype. By focusing on early-stage sketching and ideation, it contributes to Human-Computer Interaction and Design Education by redefining how generative AI can support creative design workflows.

This is through pivoting the usage of the AI tool as a support tool instead of a production tool. Rather than functioning as an autonomous producer, the AI tool is positioned as a collaborative pedagogical partner and as a mediator for cognitive assistance with depicting the user's conceptualizations onto a digital canvas [8, 5].

The project also contributes to a niche application of embedding reduced-fidelity AI suggestions directly into the workspace as a reference layer. These systems create a supportive feedback loop that helps advance designers' "initial models" into more developed "synthetic models" [33]. More broadly, by exploring AI technologies for creative practice, this project engages in a meta-reflective [20] reflective conversation about how AI can benefit human action, rather than overtake them during ideation, the development of creative or technical design artifacts, as well as the underlying cognitive capabilities that constitute authorship and creative agency.

Literature Review:

Sketching is an essential and fundamental human activity for communicating ideas and bridging linguistic borders [35]. Sketching can be considered one of the most versatile tools for externalizing ideas for visual communication across various disciplines ranging from art to design to engineering to name a few [32]. Sketches are also fundamental within the field of Design as they are visual representations of an individual's conceptual framework of their thought process.

In the early stages of the design process, designers utilize freehand sketching to explore possibilities through generating different concepts that can later be refined for final design decisions [19]. The unrestricted and ambiguous nature of sketching enables creative thinking since it is not constrained by the know-how mechanics of most computer-aided design (CAD) tools [19]. This means that the act of sketching is not just a motor skill but a profound cognitive activity where designers identify features and concepts for larger goals [19]. This translates into how designers can then share and express their ideas (mental models) to others via their sketches and written/verbal communication skills.

The recent addition of touchscreen devices such as digital tablets has made sketching more accessible due to the software applications [35]. These new technologies have enabled designers greater access to being able to showcase and communicate the user's visual ideas of their product designs to others more easily. There has been support and development in Artificial Intelligence (AI) (such as deep learning and generative models) that has now opened new opportunities for human-computer interactions [32]. These AI systems are software based and have the potential to support and improve a Designer's workflow [32]. This literature review will explore the three core areas relevant to the implementation of an AI-assisted sketching:

1. Sketching and Learning

2. Assistive Technologies and Cognitive Impact
3. AI Usage as Pedagogical Tools

This Literature Review identifies gaps in current research regarding the integration of generative AI as a non-intrusive reference layer. Rather than automating the final output, the system provides a digital underlay that supports and scaffolds the designer's creative process.

1. Sketching and Learning:

This section establishes the theoretical foundation for the project by examining sketching as a cognitive, pedagogical, and communicative practice within design. This section argues that sketching is not just a means of representing ideas, but a process through which designers construct, externalize, and revise their mental models over time. Subsection 1.1 considers the role mental models and schemas have on shaping how designers interpret and communicate their internal concepts into visual form. This section will also cover conceptual changes and theory revision to show how new information can prompt the reworking of existing understandings and the development of more refined or synthetic models (Subsection 1.2). Subsection 1.3 discusses sketching as a form of visual thinking and externalization and its iterative design process. Section 1.4 addresses skill building from novice to expert, highlighting how sketching develops not only as a technical ability but also as a mode of thinking that can be supported through pedagogically assistive systems. These perspectives provide the conceptual grounding for the proposed AI-assisted sketching framework and clarify why generative AI should be positioned not as a replacement for human design activity, but as a scaffold that supports the development and communication of designers' internal mental models.

1.1. Construction of Mental Models:

Sketching plays an important cognitive role within the design process because it allows individuals to externalize and refine their ideas (mental models) all the while learning throughout the process [12]. The Theory of Mental Models revolves around how individuals do not just interact with the world directly but through their own internal representations of the world [12]. Johnson-Laird defines mental models as structural representations that mirror aspects from the world that allow individuals to reason with spatial and conceptual information that go beyond verbal or written description alone [12]. This difference is important to designers as sketching is an external activity that represents their internal mental processes. While the designer sketches they can interact with their constructed sketches through imagery rather than just abstract descriptions (usually in the form of text) and this can make their thoughts visible and open to revision [12].

The construction of mental models is not based on a random process but rather is built upon schemas. Rumelhart [25] chose the term schema (or schemata) as the “building blocks of cognition” [25]. The reasoning was that schemata are fundamental units that are used to represent knowledge in memory to provide an outline

for how new information can be processed and interpreted [2]. With regards to the act of sketching the designer can categorize schematas based on their features so that they can “fill in” missing parts of mental representation based on their prior experience [2]. This means that a designer can then utilize imagery that they have already seen/experienced to elaborate and build upon their working designs. And at the same time while the designer is comprehending their ideas they include a mental description of the idea/project. This internal information is then used to construct a mental model that represents the designer’s intentions [14, 2].

Mental models are also shaped and refined throughout the design process [2]. This is based on how a designer experiences different schematas that can then impact the formation of their mental models. As the designer has their own internal mental models formed around their design goal, it is still subject to being influenced throughout the process [2].

There are three distinct kinds of learning that can influence a designer’s mental model: Accretion, Tuning and Restructuring [2]. With regards to how this project handles an iterative sequential usage of AI-in-the-Loop for design workflows, the most impactful learning style would be Tuning. Tuning involves the slow modification and refinement of a schemata through the individual's experience [2]. And because sketching is sequential, there's room for refinement and modification in the way that the Designer draws their mental models.

While Tuning explains how a designer gradually refines an existing idea through experience, it does not fully account for moments when new information prompts for more substantial reorganization/reflection [2]. Vosniadou makes an account of how mental models become especially useful: “mental models are dynamic structures that are created ad hoc or on the spot”. This is because they state that mental models are formed for the specific purpose of problem solving by providing a reasoning for an answer to a problem [34]. These mental models are formed and constrained by the designers’ [34] underlying conceptual structures based on their prior understanding of the world (such as the laws of physics). When the designer is presented with new, sometimes conflicting, information they must reconfigure their mental models. Vosniadou describes this process as the construction of “synthetic models”. This happens when the individual tries to merge their “initial mental models” (based on prior data) with the newly acquired information [34].

This mental constructive process should be central to how designers should implement AI tools within their workflow. The use of an AI suggestive underlay is one approach to this solution. If the designers’ mental model is constrained by their limited background knowledge then the use of Generative AI suggestions can act as a catalyst for conceptual change. By providing it as a reference layer, the system can now introduce new information that then requires the designer to review their initial concept and to consider reforming or tuning their mental model [34]. The reference layer essence becomes a “reflective conversation” that the designer is having with the AI material [12] and this can support how the user is more likely to remember their reasonings for design changes. This process is executed by the construction and testing of their mental models [12].

Drawing serves as a method to externalize and to manipulate the person's mental models [12]. Through sketching on top of an AI reference layer the designer can then "test" their own models against that of the AI system. This allows the designer to make sure that their final design retains relevance to their initial goal while also expanding on their creative design solutions [12]. This supports that AI implementation should not just automate to a final output, but rather provide the designer with a scaffolding that can aid them in developing and communicating the designs of their mental models.

An AI-assistive sketching layer is used to generate a reference layer (based on how the designer conveys their mental model). However it is more than just a reference as it can act as a catalyst for the next iterations the design will have. The designer can undergo *Tuning* [2] as they begin to slowly refine their current mental model and potentially the Restructuring learning style can be applicable if they form new concepts based on the information received.

1.2 Conceptual Change and Theory Revision:

Conceptual Change happens whenever there is a revision or any reconstructing of pre-existing ideas/concepts [34]. This reconstruction can occur when the designer's initial mental model (which is based on their previous knowledge) encounters new additional (and maybe conflicting) information. Vosniadou [34] credits this as a movement from the "initial models" to "scientific models" through a series of "synthetic models" [34]. These synthetic models are a designer's attempt to reconfigure their background knowledge with the new additional material. This project takes advantage of this via the use of a suggestive reference layer that is generated through AI models [34].

This conceptual change can also be attributed to how an individual handles imperfect analogies in a given scenario [26]. Usually pre-existing schemata are used to interpret a new situation and then the designer can then compare their initial mental model (Source) with the AI-generated Reference layer (Target) where they can then find features that may or may not match [26].

for conceptual change to occur for the designer, they must engage in reflecting their initial ideas. The way Vosniadou [15] went about her study was through the people's observations of astronomy. She had found that children often believe that the ground is flat because it appears to be flat, however they adopt a spherical model once they recognize that flatness is only a local perception of being on a larger sphere. The inclusion of a reference layer can then act as another perception for the designer to experience conceptual change. The designer's mental model serves as a [34] outline that can then allow them to generate predictions and explanations for how the design works. These models need to be externalized through sketching so that the designer can recognize if there are any inconsistencies between their sketch and their internal ideas [34].

Restructuring learning can be further explained through the notion of “Conceptual Fluidity” and “Conceptual Slippage” [7]. Hofstadter claims that concepts are not just fixed entities but rather are susceptible to change. Whenever the mind has to apply any pre-existing concepts to a new context, that concept can be changed to fit within the new context [7]. Basically, “cognition is recognition” and any analogical thinking (making analogies from one entity to another) can alter the initial idea. This process is analogous to the field of science where old standards are discarded in favor of new findings [7].

This project's multimodal AI Sketching system produces a resulting reference layer that can then enable “slippage” by allowing the designer's initial concept to change. Unlike other implemented AI tools for design workflows that focus on directly influencing the final design output, this provides a fluid architecture for [7] the creative construction of new concepts. This is because the designer is able to perceive and internalize the key features from the reference layer and make design changes. This exposure is critical for avoiding Design Fixation so that the designer can treat the AI not as a final output tool but as an example that can be referred to [34].

1.3 Visual Thinking and Externalization:

Sketching can serve as a critical tool for externalizing internal mental thoughts into a medium that can be communicated to others [30, 32]. The process of sketching is an act of “visual thinking” that the designer has to comprehend and inform and understand with visuals rather than relying on just language [32]. Designers depict their internal thoughts into sketches to act as a “literal model of an idea” that allows for the designer to check if their concepts are consistent [32]. A primary function for this externalization of their sketch is because it can be considered as cognitive offloading. This is due to how the sketches serve as an external representation of their ideas and thus the memory and information processing that is done internally through the mind doesn't need to be called upon consistently [32]. Thus the designer's mind is relieved of a great burden when trying to recall their ideas/concepts as they can refer to their sketches. And of course the act of sketching is critical to students to externalize their thoughts for problem solving [19]. Even a sketch that is undergoing work can be considered a “reflective conversation” as the drawing is developing to fit new ideas [32].

Sketches, for better or worse, are not highly rendered and thus are characterized by their [32] abstract and illustrative nature. This is due to how they consist of fewer strokes, and other details that are omitted during the early design phase. Sketches can be understood as schemas based on reality [27]. Because the sketches omit higher level detailed information they consist of predominantly necessary features that are critical to the design [32]. This is what makes sketching a powerful tool as it can distill ideas in simpler form [32].

Sketches can have a dynamic and sequential nature to them due to how the designer externalizes them. This is because humans can only draw one stroke at a time (with the exception of using both hands which is unlikely). The iterative process allows for continuous visual feedback and the designer can make adaptations and changes to their design as they sketch [32]. This project software is meant to provide AI assistance via generative imagery. By providing a “suggestive scaffold”, this system can introduce additional visual cues that can be interpreted by the designer. And even further understanding this process can justify the use of an AI underlay as a partner in the development of the designer's mental model [19, 32].

1.4 Skill Building (Novice to Expert):

The difference between a novice designer's sketching capability and an expert involves not only an improvement in their motor skills but in how they utilize sketching as a cognitive tool. Novice designers for instance face challenges such as accuracy, perspective, line quality and more [19]. Merzdorf notes that while sketching has been taught using traditional methods the focus is now shifting towards innovative problem solving [19]. This shift is critical since experts utilize “thinking sketches” as a way to facilitate and explore design solutions [19].

There has been an increase in the development of computational assistive tools that are meant to aid designers who may need assistance or guidance during the design process. Shadowdraw is a clear example where it was developed specifically to assist the user with little intervention as it would provide a shadow-based suggestive feedback [17]. User studies reveal that such AI guidance tools are most effective for users who possess [17] little background in drawing skills. While experts find shadows distracting and novices struggle with initial strokes, “average” users on the other hand benefit significantly from AI assistance [17] in realistic proportions and structure [17].

Generative AI tools such as “Inkspire” address the specific prompting limitations often faced by novice users [18]. These novice users often struggle with limited prompting capabilities and often just copy keywords from a brief. By providing an [18] visual inspiration and converting the high-fidelity renders into “abstracted set scaffolds”, these tools lower the cognitive barrier for novices. These scaffolds allow the user to build upon the suggestions without becoming fixated on rendering the final output [18]. This project is meant to enable a skill building AI assisted system that encourages a sketch-design-sketch feedback loop [18]. This is because as new designers learn how to sketch the AI can act as a support tool and as a co-creative design partner [18]. Utilizing intelligent digital tutors with adaptive feedback aids designers in mastering their “visual language” so that they can properly communicate properties of their designs with others [17, 19]. This can be considered a pedagogical approach that ensures AI is a supportive tool for the natural rhythm of interaction that guides them into depicting their desired mental models to others [18].

2. Assistive Technologies and Cognitive Impact:

Building off of the prior sketching section of this paper this next section emphasizes the impact of technological interactions that designers can be cognitively impacted by. More specifically the topic of Activity Theory (AT) is a significant factor in this project as it is used to analyze interactions between technology and humans [11]. This is supported by the Russian social psychologist Vygotsky and incorporates activity systems that revolve around the individual with their community and tools and how it plays into their objectives [11].

2.1 Mediation and Activity Theory:

The core principle of Activity Theory is that humans form relationships with the environment (the world) however it is mediated through culture such as tools, language and socialization [5,11]. This theory is integral to this project as it revolves around implementing an AI program that can mediate between the designer (Subject) and their mental model (Object) [5, 11]. This mediation is not without its after effects as it can carry “historical and cultural baggage” that can impact how the designer interprets the project and interacts with it [11].

The activity system provides a model to understand how the actions of the designer (the Subject) are governed by the social rules and community [5]. Their mental model (the Object) is represented as the “problem space” where it can be impacted by both physical tools (a digital tablet and software) and symbolic signs (text prompts and visual references) [5]. With regards to the designer the AI’s suggested reference layer acts as a mediate tool that allows them to change or develop the “difference of potential” of their initial mental models with the suggestions made by the AI’s multimodal system [11].

Mediating is implemented in a dynamic and evolving manner. This is because the process can serve as a “re-tooling or re-mediation” where when there are different subjects (different users) they may utilize the same technology with varied intentions and goals [11]. This project chooses to not utilize the AI program as just a tool to automate an output but to provide a “reflective conversation”. This “reflective conversation” consists of the designer and their tools and how they can “re-mediate” their design based on the system's feedback [11]. Taking into account that designing can be an intentional activity the use of AI suggestive tools can facilitate a potential outcome for the final sketch. The inclusion of the designer, AI tools, and the working sketch implies that this is an interdependent system [11]. And again the AI suggestive tools serve a purpose of mediating between the designer and their sketch based on their intentional design [5].

2.2 Distributed Cognition:

Distributed Cognition is the phenomena of properties that are within an interdependent web involving the brain body and world/environment [9]. With respect to the designer in this project, the AI program is not entirely a separate entity but instead is considered to be part of the shared environment (the software). And with this in

mind, the final sketch outcome can be viewed as an emergent property [8] because it comes from a system based on interactions. This means that the creative outcome is not solely based on the designers intent not the AI's algorithm but instead it is derived from the relationship based on the designers sketching strokes and the AI's suggestive layer [8].

A generative AI layer can act as a functional organizational tool based on the designer's cognitive capabilities. The program for this project allows for the designer to offload the burden of retaining complex spatial concepts (their mental model) within their working memory through the use of an AI reference layer. This alludes to what Hutchens [8] would describe as an "enacted reasoning" where the act of manipulating the physical work is essentially a process for thinking [8, 9]. This interaction relies on the coordination between internal and external structures [9]. The AI's reference layer can act as an "external representation" that can provide stability and guidance for higher level reasoning with the design. The designer can take references and internalize and refine them through "reflective conversations" with the sketch through their internal mental understanding [8, 25].

If implemented correctly the use of an assistive sketching tool should qualify for enabling the designer to be in a state of "readiness-to-hand". This means that drawing with the tool becomes familiar enough to the user that they no longer require conscious awareness of the tool. On the contrary if the tool makes the user (designer in this case) constantly conscious of it, then that means it's still "present-at-hand" [21]. Within this project, the use of an AI underlay that is within the background and only generated when demanded means it could aim for a "seamless coupling" which means it could be absorbed within the designer's workflow and an embodied interaction [21].

2.3 Framing, Keying and Social Interaction:

To understand how the designer interacts with references, we have to look into the concept of "Frame Analysis" by Erving Goffman [3]. This implies that individuals can organize their experiences to make sense of a situation such as "what's going on?" [22]. Within the context of this project, the AI-assisted sketching tool creates an environment that can answer the question that the user has not considered yet: "what is being conceptualized?". The AI suggestions can transform the designers perception of their current state of the sketch and those the designers perception of their mental model can be speculated [22, 3].

This interaction is built upon the concept of "Primary Frameworks" [3] which can be considered "cognitive building-blocks" of the individual's experience. The start of any sketch is a blank canvas however when the designer has any relevant visual data before them they can formulate a framework for the design. An AI underlay can trigger "transformations" [22].

2.4 Affordances and Signifiers:

2.4.1: Foundational Concepts of Affordances:

This next section is arguably one of the most explored and impactful topics with regards to design. This is because it is about the relationship between the Designer and their designs that consists of affordances and signifiers. These are a multitude of available actions that the designer can have when interacting with their environment and objects. The theory of affordances was pioneered by James Gibson [31] who expresses that the environment offers/contains certain behavioral characteristics [31]. To be more specific an affordance is an ecological property [31] that refers to the environment and how the person interacts with it. But to be an effective affordance it must be measured with respect to the person within the environment (where in the case of this project would be the digital workspace) would interact with it [31]. This implies that the sketching tool and sketching environment only affords the creative exploration for the designer if it contains features that can accommodate the designer's cognitive capabilities [31]. Turvey (1992) took this further by formalizing how the characteristics of affordances act as properties which are essential possibilities that can exist even if they are not used [31].

Since this project consists of AI-assisted sketching tools it is subject to influence how the designer perceives their canvas differently as it is no longer just a blank, non-responsive canvas. It can depict potential creative avenues however they are only suggestions made by the AI system. And because it is a reference layer it depicts potential (latent) affordances that are waiting to be activated by the designer when and if they include those features in their working layer.

2.4.2: Signifiers and their Cues for Discovery:

Now while affordances may represent the potential actions that are capable in an environment, a signifier differs as it is a perceived signal that makes the affordance more noticeable [31]. Norman [25, 26] had mentioned that signifiers are essential in increasing interaction efficiently because they can indicate higher level information for important affordances [31]. Signifiers are basic concepts created with consideration of an environment influenced by an agent, as noted by Vachtsevanou [31]. Now with this in mind signifiers can enable the designer to proactively discover how to interact with the environment [31].

And to again bring this back in relation to the project, this means that suggestive reference layer serves as a primary signifier as it is a perceivable cue/hint for what to draw/sketch next [31]. By treating the AI reference layer as a signifier the system doesn't explicitly dictate how the final outcome will be and thus retains the intentionality of the design to how the designer sees fit [31].

2.4.3: Embodied Interaction and Schemas:

Affordances and signifiers can be more impactful when they are utilized together through an embodied interaction. This means that when a tool becomes familiar, it can be an extension of the user's body and a non-conscious effort for a call to action [2]. A real life analogy would be such as a blind man's stick can no longer be considered just an object instead an instrument for perception [2]. And for this project it translates into providing an integrated sketching tool that assists the designer to focus more on conveying their mental models even if they lack the skills.

2.6 Design Fixation (aka: The “Cognitive Risk”):

Design fixation represents a significant "cognitive risk" when it includes the integration of Generative AI (GenAI) into the creative process. This can happen when a designer's exploration of their design is restricted due to a potential unconscious bias towards existing examples or prior knowledge [3]. While this area has traditionally been studied through the human cognitive, recent research suggests that GenAI models themselves experience a parallel form of design fixation (which is not surprising due to the fact that they need to be trained on pre-existing datasets), which can lead to similar outputs and an overall reduction in creative diversity [3].

2.6.1: Technical and Human Causes of Fixation

The causes for this potential cognitive risk can stem from the complexities of human-AI interaction. There's a technical aspect that presents significant challenges such as with data training limitations. Unbalanced datasets that contain prejudiced information can cause models to learn and perpetuate biases, resulting in a lack of diversity in style or concept [3]. There are also architectural limitations that influence how the inputs are processed and this can fundamentally shape the potential types of outputs. And of course human factors also play a role as fixation can come from lack of clear input such as prompt writing or unclear imagery. Novice designers in particular might struggle with limited prompting capabilities and this can lead them to copy keywords from their design briefs rather than exploring their own design spaces [18].

2.6.2: Mitigation through Scaffolding and Analogies

To combat cognitive risks there is current research that promotes the use of lower-fidelity assets and multimodal interfaces. Such tools like Inkspire address fixation by converting high-fidelity AI designs into abstracted "sketch scaffolds" [18]. These scaffolds are presented as a transparent underlay beneath the designer's canvas and act like tracing paper to help users build upon AI designs without being distracted or fixated by photorealistic details [18].

Analogy driven design utilizes concepts from the designers environment and encourages them to move beyond a single thread of thinking [18]. By integrating these reference layer underlays and diversifying the source of input (text and images) an AI-assisted tool can transition from just providing outputs to being an assistive tool

that enables the designer to build upon what they are capable of sketching and what their mental model is perceived as [3, 18].

2.6.4 Current AI Sketching Tools and their Interactions

The integration of AI within creative workflows is a growing field. One implementation of AI as a supportive tool for creative use is as an Intelligent tutoring system. These Intelligent tutoring systems have been developed to provide personalized feedback and guided practice to help users improve their technical sketching skills . These systems can either make use of sketch recognition algorithms to interpret student drawings and provide feedback on aspects like accuracy and perspective and even can provide specific guided tutorials for users.

AI technologies have now advanced to a point that they can now be utilized and explored as a collaborative partner within the creative design process. The ShadowDraw project demonstrates that a large language model (LLM) can be prompted to generate sketches sequentially in a way that can be collaborative alongside human users [32]. This is achieved by defining an intuitive "sketching language" that the LLM uses to specify strokes on a grid-based canvas [32]. This work highlights a critical shift from AI as a tool for generating a final product to AI to instead an interactive agent that participates as a co-designer alongside human use cases [32]. This concept of human-AI co-creation is what started this project's initiative and the core for how to improve on its implementation in the design field. This niche application of ai technologies assisting the user without taking authorship of the creative design decisions of the designer is significant in retaining the intentionality of expressing the designers' mental models to others.

3. AI usage as Pedagogical Tools: Human-AI Collaboration in Design and Learning

The introduction of Generative Artificial Intelligence (GAI for short) tools within design education has the potential to reshape traditional learning methods. One case where this can be implemented is within Bloom's Taxonomy [24]. A case study involved students who were tasked with designing a robot and they had access to GAI tools. This study found that AI's role within a problem solving process changed depending on the students' familiarity with the task at hand. When students were knowledgeable about the topic (such as sketching or CAD), the AI tool served as an accelerator that enhanced their ability to analyze, evaluate and even problem solve [24]. However, when faced with an unfamiliar domain they were not so knowledgeable about (for example: programming an Arduino), the students' use of GAI tools was to directly create solutions before they even fully understood the underlying problem(s) [24]. This suggests that Generative AI can enable incorrect learning where the practical application to problem solve instead precedes theoretical understanding of the problem [24].

This showcases a valuable reason for designing an AI-assisted sketching tool. The AI tool should act as a scaffold that enables users to explore complex design variations they might not have the technical skill to produce on their own [24, 36]. Circling back to the robot design project as an example, the GAI tools were used for brainstorming initial concepts as well as adding depth and detail for the visual representations and renderings [24]. However obtaining those skills to begin with requires critical thinking and judgment from the user as that is how they can evaluate the AI's output [24]. Basically students' ability to be critical when evaluating GAI outputs varied based on their prior knowledge and understanding of what they were solving for. There could be instances where they may heuristically discover how to utilize GAI tools more efficiently based on their prior background knowledge. When the User of a GAI tool is unaware of the subject they don't have any reliable heuristics that can enable them to understand how to offload their cognitive capabilities onto another system.

This again highlights the core motivation for this project: creating a tool where the AI provides suggestions as an underlay. However the "suggestion in question is an interpretation of the user's mental model. The ai usage is limited to only an underly because this interaction can keep the human designer firmly in control of the design process. This allows for only using the AI's output as reference material or as a suggestion rather than asking for a final solution. It also stays relevant to the traditional methods such as using tracing paper when creating visual design and using an underlay to work off of. The user can retain more authorship as they are the one making conscious decisions about whether or not to take into consideration the AI's output or to refine an assisted mental model. This non-intrusive approach could potentially foster a healthier and more ethical collaborative and even educational partnership that can utilize new technologies while at the same time preserving creative ownership.

3.1 Scaffolding, Templates, and Inscriptions:

The implementation of an AI-assisted sketching system can transform and build upon the traditional creative workflow into an assistive-creative process where the AI acts as a scaffolding partner providing support through underlays. This relationship is best understood through the theoretical lens of templates and inscriptions, which describe how knowledge and agency are embedded into technical artifacts to guide human behavior. In this context, the AI does not merely produce an output but provides a generative template that serves as a building block for the construction of a design [16]. As a generative principle, a template orientates sense-making and prescribes a behavioral pattern across a variety of media, ensuring that a given stock of knowledge such as a specific design concept is durable and transferable [16].

This project highlights the process of inscription and delegation. Inscription is a form of "authoring" where a program of action is written into a stable medium to become both the receiver and the carrier of human activity [16]. When a designer uses a text prompt to generate an underlay, they are delegating certain cognitive and motor tasks to the AI. Essentially they are authorizing the system to take over and represent their initial design

intention [16]. By delegating the creation of a base silhouette or structural proportion to the AI, the designer is relieved of the immediate burden of "thinking through" the blank canvas, allowing for greater cognitive efficiency in the early ideation phase [16].

This ability depends heavily on the fidelity of the sketch. A central challenge identified in current generative AI research is that AI-generated designs often appear "too complete," which can induce design fixation by making it difficult for designers to envision new ways to build upon or iterate the work [6]. To mitigate this "cognitive risk," the AI implemented in the system utilizes a mechanism of sketch scaffolding as a way to convert high-fidelity AI renders into a lower resolution abstract version of the sketch underlay [6]. This "scaffold" acts similarly to "tracing paper," providing a transparent view into the current state of the AI's internal representation without the details of a photorealistic image [6].

This "reduced-fidelity" approach is grounded in the principle that ambiguous sketch underlays reduce fixation on conventional forms and promote divergent thinking [6]. By transforming a "too complete" image back into a loose, abstract abstraction, the system completes a "sketch-to-design-to-sketch" feedback loop [6]. In this loop, the scaffold serves as more than just a reference point; it becomes a shared mental model and a communication tool between the designer and the AI partner [6]. The designer can then "steer" the AI by either tracing over or intentionally deviating from the scaffold, maintaining their subjective agency while benefiting from the AI's generative capacity [6].

Finally, the success of these scaffolds as pedagogical tools relies on their ability to exploit the intrinsic traits of free-hand sketching. Sketches are naturally highly illustrative yet concise and abstract, reflecting the "brain's internal representation of the world" [35]. Because they lack background clutter and utilize iconic representation, these scaffolds benefit from automated analysis and real-time interaction in a way that dense photographs cannot [35]. By inscribing these abstract "templates" as a layer beneath the working canvas, your tool provides the necessary scaffolding for a designer to progressively refine their own mental models, transforming the act of sketching into a truly collaborative "reflective conversation" between human and machine [6].

3.2 Generative Collaboration and AI Fixation:

The shift from viewing artificial intelligence as an autonomous generator to a collaborative partner emphasizes the importance of how human agency and intention are preserved in co-creative contexts. Generative collaboration is not merely the production of an end-product but a dynamic interaction where the AI serves as a "meaning machine" that provides signifiers (albeit a bit randomized due to AI) the human designer can then interpret new meaning [6]. However, this partnership is complicated because of the phenomenon of GenAI design fixation. This happens when the AI model restricts its exploration of the generative space due to "unconscious biases" that are rooted within its training data and architectural constraints [3].

3.2.1 The Nature of Generative Collaboration

True collaboration requires the creation and consumption of meaningful context, connection, and interpretation which are elements that Generative AI fundamentally lacks [6]. Harwood [6] characterizes AI-generated content as if it were produced without regard for truth or inherent meaning, functioning instead as a "pattern visualization machine" that mimics human thought [6]. Despite this however AI can become an effective interface for capturing and representing creative intentions if it is utilized for human reflection [6].

To move beyond the "slot machine" interaction of one-step generation, recent advancements like SketchAgent emphasize the sequential nature of sketching [32]. By drawing stroke-by-stroke using string-based actions on a grid canvas, their agents capture the evolving and dynamic qualities of human ideation [32]. This allows for interactive human-agent sessions where both parties contribute meaning towards the final output, achieving recognition rates close to those of solo human designers [32]. In this collaborative mode, the AI takes on specialized roles: the Puller (gathering context), the Pusher (offering suggestions/variations), and the Producer (creating specific artifacts like components rather than full works) [6].

3.2.2 The Cognitive Risk: Fixation in Collaboration

The primary obstacle to effective collaboration is the risk of fixation. GenAI design fixation manifests as homogenized outputs that are limited to a certain amount of variation, and a susceptibility to high-frequency words (such as "ergonomic" or "sustainable") in text generation [3]. In image generation, this fixation appears as a limited range of shooting angles, color, shape, and textured attributes [3]. These technical fixations can limit the user's design fixation, since they might find AI outputs "too complete," making it difficult to envision alternative versions of their design and creativity over the long term [3].

3.2.3 Mitigation and Empowering Human Agency

Mitigating these risks requires a hybrid workflow that prioritizes human authorship and intentionality. Chen [3] suggests that rather than providing direct answers, collaborative tools should provide "situation-based stimuli" that can instead guide designers to think more critically [3]. Strategies to alleviate fixation include:

- Reduced Fidelity: Utilizing low-resolution underlays to leave room for exploration [3].
- Multi-Agent and Human-AI Collaboration: Endowing AI with critical thinking through multimodal systems and allowing humans to refine and expand upon AI-generated content [3].

The goal of generative collaboration is to ensure that AI compliments rather than replaces the designer. While AI can analyze composition and/or mimic style, it cannot understand why a design choice is made [6]. By assigning tasks that require judgment, intuition, and emotional intelligence to the User, the more repetitive tasks can be offloaded to the AI the system to help the user with their creative goals throughout the design process [6]. This balance ensures that the final design remains relevant to the designer's intentionality and is built upon the decisions made by them, rather than based on the visualization by the AI [6].

3.3 Scaffolding and Analogical Reasoning:

The success of AI as a suggestive tool relies on its ability to bridge the gap between abstract design intentions and the visual form of the design. This transition is cognitively bridged with analogical reasoning which is a process where new knowledge structures are created by modeling them off of existing, familiar schemata [26]. In the context of AI-assisted sketching, this involves a systematic mapping of prior knowledge from one domain to the target design problem to facilitate restructuring or even the creation of new mental models [26] [18].

The mechanism for learning by analogy involves taking a pre-existing schema and using it as a way to create a new one that is similar but different in specific modifications [26]. Any characteristics of the old schema are not completely different from that of the new version because they are similar in conception and relationship. This is important to the designer as they can carry over features and make rapid iterations of their design without needing completely new resources to base their new versions from [26].

For a pedagogical tool to be effective it should provide a conceptual model based on what the designer is already knowledgeable of. This means that the target and source domains differ by a minimum number of dimensions [26]. This theoretical grounding justifies the use of visually-concrete inspirations to assist designers who often struggle to convey abstract concepts through text alone [18].

Recent advancements in co-creative tools like Inkspire (6) that utilizes Large Language Models (LLMs) to generate analogical anchors [6]. For example, an abstract design theme such as "protectiveness" is grounded through analogical reasoning into concrete objects like a "tortoise" or a "shield" [18]. This happens when parallel relations from a source domain (e.g., a tortoise's shell) are applied to a target domain (e.g., a car's exterior) even when some features differ [6, 20]. By recommending analogies (such as from nature, fashion, or architecture) the AI acts as a "meaning machine" to help the designer recognize creative possibilities without the cognitive friction of manual prompt engineering [18].

Furthermore, analogical reasoning serves as a critical countermeasure to GenAI design fixation [3]. Chen notes that generative models are often susceptible to "repetitive themes" and "limited contextual variation,". This means they frequently default to conventional patterns found in their training data [3]. Utilizing the design-by-analogy method diversifies the designer's thinking and expands the range of solutions considered, effectively "breaking" this fixation of homogenized/similar AI outputs [3].

The implementation of sketch scaffolding functions as the physical manifestation of this analogical reasoning. By abstracting high-fidelity AI designs into low-fidelity, "loose and abstract" underlays, the system provides a

transparent view into the AI's internal state [18]. These scaffolds act as "tracing paper," allowing the designer to engage in a sequential dialogue where they can either trace over or intentionally deviate from the AI's suggestion [18]. This completes a sketch-to-design-to-sketch feedback loop, transforming the interaction into a shared mental model where the designer and AI partner to refine concepts iteratively [18]. Ultimately, this scaffolding ensures that the AI supplements human intent by identifying conceptual gaps rather than dictating the final stroke [18].

3.4 Intelligent Tutoring and Feedback Loops:

The integration of artificial intelligence into sketching environments facilitates a transition from passive digital canvases to active intelligent tutoring systems (ITS) that provide adaptive, personalized guidance. As classroom sizes grow, the demand for scalable pedagogical support has led to the development of intelligent software that can guide students to master sketching through AI-based instruction [19]. These systems translate complex sketch recognition algorithms into actionable instructional methods, such as guided practice and real-time feedback, allowing learners to develop motor and perceptual skills at their own pace [19]. Within this framework, the AI functions not as an automated producer of art, but as a co-creative partner that mimics the presence of a teacher to provide scaffolding throughout the drawing process [17].

A primary mechanism for this pedagogical intervention is the real-time guidance system, which provides immediate visual feedback as the designer works. ShadowDraw exemplifies this approach by dynamically updating a "shadow image" underneath the user's strokes, suggesting object contours based on local edge patches matched from a massive image database [17]. By providing these suggestive references, the system functions like a partner that "enlightens the artist" with the gist of multiple visual references simultaneously, rather than forcing the user to trace a single static image [17]. This form of interaction preserves the user's freedom and expressiveness while ensuring that overall shapes and proportions remain realistic, a critical factor for building the confidence of "average" or untrained sketches [17].

The effectiveness of these tutoring systems is further enhanced by the implementation of a complete sketch-to-design-to-sketch feedback loop. Recent advancements in tools like Inkspire emphasize that AI should generate a new design iteration every time a single pen stroke is drawn, thereby encouraging designers to consider small but meaningful changes in form with each addition [18]. This near real-time performance creates a positive feedback loop where the user receives suggestions while still in the process of externalizing their thoughts [17]. Crucially, the system converts these high-fidelity AI generations back into low-resolution sketch scaffolds, serving as a "transparent view" of the AI's current state that the designer can then build upon or intentionally deviate from [18]. This iterative cycle ensures that the designer remains engaged in a "reflective conversation" with their material, utilizing the AI suggestion as a control point to plan subsequent strokes [18, 19].

Finally, the pedagogical value of these systems lies in their ability to capture the sequential and dynamic nature of human drawing. Humans do not create designs in a single step; they draw iteratively, stroke by stroke, incorporating continuous visual feedback [32]. Agents such as SketchAgent leverage the prior knowledge of multimodal LLMs to engage in dialogue-driven, sequential sketching, where the agent can pause at specific tokens to allow the human user to add their own strokes to a shared canvas [32]. By integrating user-drawn strokes back into the model's sequence, the AI facilitates a collaborative turn-taking process that mimics a social-cognitive interaction between a student and a mentor [32]. This sequential approach ensures that both the human and the AI contribute meaningfully to the recognizability of the final sketch, effectively training the designer to master the "visual language" necessary for communicating complex physical and functional properties [19].

Methodology

As presented in the introduction and related work, the conceptual framework for this project centers on three primary themes: (1) Sketching and Learning, which examines sketching as a cognitive and pedagogical practice for externalizing and refining mental models [12, 19]; (2) Assistive Technologies and Cognitive Impact, which explores how assistive systems influence cognition, agency, and creative ownership/authorship [8]; and (3) AI as a Pedagogical Tool, which positions AI as an active partner in design education [19]. Figure h presents a visualization of this framework.

The framework illustrates how these three themes converge within the MSM sketching canvas and inform the Sketch-Convey-Reflect loop. Sketching and Learning contributes the role of sketching as a method for externalizing mental models; Assistive Technologies and Cognitive Impact addresses how AI systems influence cognition, authorship, and user agency during interaction; and AI as a Pedagogical Tool frames generative AI as an active collaborative partner within the design process. Together, these themes shape the development of the MSM system as an AI-assisted environment for ideation and reflective design practice.

The lower section of the framework represents the intended outcomes of this interaction model: AI-enabled support, creative agency, and reflective practice. These outcomes guided the development and evaluation of the prototype throughout the iterative Research-through-Design process.

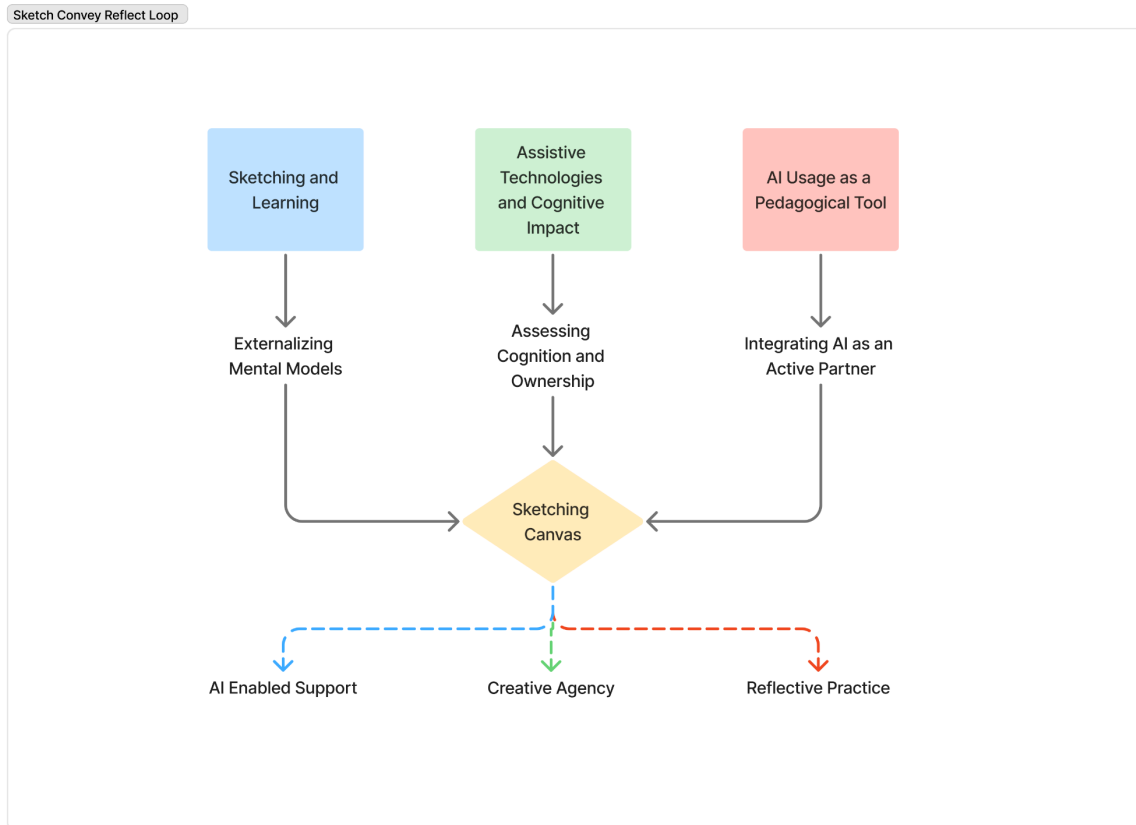


Figure A.1: The diagram showcases how the interactive sketching canvas is central in incorporating the three core principles from this project to its desired goals

This framework also informs the project's central research question:

How can AI technology be utilized and interacted with by the user in ways that preserve their intent and creative agency while providing assistance during the sketching process?

From this framework, three research questions were developed focusing on the relationship between AI-assisted sketching, cognitive offloading, and reflective design interaction.

These themes informed three research questions focused on the technical and functional viability of the Mental Sketching Model (MSM):

- **RQ1: Sketching and Learning**

How can a multimodal AI system interpret a designer's spatial input and textual context to produce a generative reference image model that accurately grounds its output to the designer's intentions?

- **RQ2: Assistive Technologies and Cognitive Impact**

In what ways does the layered architecture and deliberate friction of the MSM interface influence the designers ability to maintain their agency and to avoid any design fixation?

- **RQ3: AI as a Pedagogical Tool: Human-AI Collaboration in Design and Learning**

How does the implementation of a non-intrusive AI-in-the-loop interaction model support the iterative cycles of externalization and reflection required within design education?

To address these questions, this project follows a research-through-design (RtD) methodology [2] that uses the iterative development of the MSM prototype to investigate how generative AI can support sketching, ideation, and the externalization of designers' mental models [18]. Research through design allows knowledge to be generated through the creation, refinement, and evaluation of design artifacts, making it particularly suitable for exploring emerging technologies and user interactions that are not yet well understood [17].

Data Collection and Analysis

Data collection in this study occurred over an eight-month period from 2025 to 2026 through the iterative development of the MSM prototype. Consistent with a Research-through-Design methodology, the prototype itself functioned as both the research artifact and the primary source of data. Data was collected during repeated cycles of designing, building, testing, and refining the system. These cycles produced multiple forms of data, including interface sketches, prototype versions, system architecture decisions, AI-generated image outputs, notes on technical successes and failures, and reflective observations about how the tool supported sketching, ideation, and cognitive offloading.

Design documentation was saved onto a local computer and included screenshots, development notes, interface revisions, prototype changes, AI-generated outputs, and revision histories. These materials captured the evolution of MSM over time, including decisions related to the interface, the Sketch-Convey-Reflect loop, and the integration of the AI pipeline.

During each iteration, the MSM prototype was tested to evaluate how well it supported the Sketch-Convey-Reflect loop. This included documenting how user sketches were interpreted by the AI system, how accurately the generated outputs reflected the user's spatial input, and how the resulting images could help designers externalize or reconsider their mental models. Technical data was also collected through validation of the two-phase AI pipeline, which used Google Gemini 2.5 Flash for vision-based sketch analysis and Imagen 4 for image generation through a Node.js/Express proxy server.

Analysis was conducted retrospectively through reflective comparison across prototype iterations. Each design cycle was reviewed to identify what worked, what failed, and what needed refinement. The analysis focused on recurring patterns related to sketch interpretation, AI response quality, designer control, visual feedback,

technical limitations, successful refinements, and the tool's ability to support cognitive offloading. Rather than relying on statistical analysis, this RtD study used reflective and artifact-based analysis to generate design knowledge from the process of making and evaluating the prototype.

Through this process, MSM was analyzed not only as a technical proof of concept, but also as a design probe for understanding how generative AI can support early-stage design ideation. The findings were derived from the evolution of the artifact, the decisions made during development, and the observed relationship between sketch input, AI interpretation, and generated visual output.

Findings

This section presents the findings and design outcomes of the project. These findings emerged through iterative prototyping, system development, and proof of concept, and include both conceptual insights and implemented design decisions. In addition to identifying key requirements for assistive AI sketching, the project produced a working prototype platform in order to reflect the research goals of supporting mental model construction while preserving user agency. The evolution of the MSM application follows five major iterative design and implementation cycles employing a Making, Testing and Reflection structure to document technical and design criteria.

Iterative Design Cycle

Cycle 1: Core Engine and Accessibility

The system was developed through five major iterative design and implementation cycles. The first cycle focused on establishing the core engine foundation. This phase emphasized accessibility and “readiness-to-hand” interaction through the use of the HTML5 Canvas API to create a web-based, hardware-independent interface. This architecture supported zero-install deployment and enabled ongoing system updates within a browser-based environment. This web-based implementation also emphasises accessibility since this approach ends up reducing initial effort or cognitive strain from the user as they no longer need to worry about any form installation or their device requirements in order to use the application. Web based implementation also supports how this project can be utilized as a pedagogical tool since as long as the user's device has access to a browser, it can benefit from this tool. Only constraints that can appear would be that HTML Canvas might struggle with higher fidelity modeling. This is low impact for the project as it focuses specifically on low fidelity, rapid ideation in order to support the user with conveying their mental model rather than producing a final output. “readiness to-hand” implies that the interface is meant to disappear during the act of creative design. This is why all of the tools are set to the side and top sections of the canvas and why the AI element is within its own section.

Cycle 2: Multimodal Integration

The second design cycle focused on multimodal integration and introduced a backend proxy using Node.js/Express to manage the complex, two-phase generative feedback loop with Google Cloud models. The system accepts dual inputs: visual spatial input (sketches), representing the user's "form," and textual semantic input (text prompts), representing their intentional design goals. This architecture utilizes Gemini 2.5 Flash as a way to perform grounding through visually analyzing the rough sketch and generating an enhanced version with the aid of the descriptive text prompt. This is then synthesized by Imagen 4 to produce a stylistic and appropriate reference image model. This multimodal approach positions the AI as a collaborative "interpreter" rather than a "guessing machine" (which would be the case if it was using just images or text alone). The text prompts acting as a "semantic anchor" that aligns the AI's generative reference image with the user's intentional mental model. Ultimately, the system's ability to interpret this mental model benefits from the increased data input and this enables the user to reflect and provide either more visual or text details that can refine the output of the reference model in order to align with their initial design goals.

Cycle 3: Multi-layered Architecture

The third cycle focused on the development of a multi-layered architecture. A layered transparency approach was implemented using a three-layer canvas stack consisting of the user sketch layer, the generative AI layer, and the background canvas. This design separated the user's editable work from the AI-generated reference material to preserve authorship and reduce the risk of design fixation. Structuring the interaction in this way also supported a turn-based relationship between the user and the system. The separation of the users layer in the AI layer reinforces an authorship/ownership of the design. By keeping the user's sketch on a distinct layer, the system signals at the AI's inputs are just to be considered as suggestions to be referenced to and not to overwrite the user's work. This is supported through the fact that the AI layer functions as an underlay, such as tracing paper would be for physical sketching. That means the user has the final say with regards to the final design decisions. Attention that can emerge would be differentiating between inspiration and design fixation, which is synonymous to scaffolding versus "takeover". If the AI layer was too detailed then the user would be more inclined to just "copy" or trace the AI layer which would diverge from the designer's intentional ideation.

Cycle 4: Interaction Refinement and Sketch-Convey-Reflect

The fourth cycle focused on user interface refinement and the "Sketch-Convey-Reflect". During this phase, the system was structured as an AI-in-the-Loop [20] model to support an ongoing reflective conversation with the material. The interface was designed to allow users to either trace or modify or even deviate from the AI-generated sketch scaffold. AI interaction was limited to moments when the user explicitly prompted the system, reinforcing user control within the design process. Explicit prompting for the AI is integral to this project as the AI can only be activated when requested and thus does not impact the user's workflow. The turn

based interaction supports humans in the loop when interacting with the AI. This allows the user to evaluate their sketch.

Cycle 5: Formative System Evaluation

Following the four design and implementation cycles, the prototype was reviewed through a technical evaluation stage. This stage focused on gathering Observations that were used in part to support the prototype refinement and are treated as design considerations for the iterative design implementations.

These observations were conducted while in the process of deploying and reviewing the application, and as such suggest that the tool can be used for divergent thinking as effectively as for refinement. This is in part because the user can prompt the AI to “make sense” of an unrefined sketch and to test boundaries of that idea. While the prototype succeeds in maintaining agency through its layered architecture for the user, it can be said that future iterations could benefit from lower latency and more granular control over eye filters. The system can be perceived as a creative mirror that allows designers, particularly students or novice experts, to reflect upon the interpretations made by their sketching and descriptions. This evaluation highlights the delicate balance of scaffolding between the AI and the user. This is because the AI reflected layer if it is too polished can negate the Ability of a user to interpret its output as part of their own

MSM System Requirements

The development of MSM was guided by a set of system requirements derived from the literature review, conceptual framework, and iterative design cycles. These requirements were intended to ensure that the application functioned not only as a technical prototype, but also as a cognitively supportive and non-intrusive sketching environment. Table a outlines the major development cycles, the refinements introduced in each phase, the core observations that motivated those changes, the corresponding system requirements, and where each component is situated within the overall system architecture. Particular emphasis was placed on accessibility, multimodal interaction, user control, and the preservation of creative authorship within an AI-assisted workflow.

Cycle	Refinements	Core Observations	System Requirements	Architecture
Cycle 1: Core Engine Foundation	Developed a browser-based sketching environment using the HTML5 Canvas API. Prioritized zero-install access,	The system needed to reduce setup friction and support “readiness-to-hand” interaction so users could focus on	Accessibility; low barrier to entry; hardware independence; web-based deployment.	Frontend / HTML Canvas Engine

	hardware independence, and rapid updates.	sketching rather than tool mechanics.		
Cycle 2: Multimodal Integration	Added support for both sketch input and text prompts. Integrated a multimodal generative AI model to process visual and semantic input together.	Single-mode AI workflows were limited: text-only input could restrict visual expression, while sketch-only input could lack conceptual specificity.	Multimodal feedback loop; sketch + text input; mental model externalization; AI-assisted ideation.	Frontend Input Layer + Backend AI/API Layer
Cycle 3: Multi-Layered Canvas Architecture	Implemented a three-layer canvas stack: user sketch layer, AI-generated reference layer, and background canvas. AI output was separated from editable user work.	Layer separation helped frame AI output as a scaffold or reference rather than a replacement, supporting authorship and reducing design fixation.	Layered mental model representation; authorship preservation; transparent AI underlay; design fixation mitigation.	Frontend Layered Canvas Stack
Cycle 4: UI Refinement and AI-in-the-Loop Workflow	Refined the interface around a "Sketch-Convey-Reflect" loop. AI generation occurred only when explicitly prompted.	The workflow needed to keep AI reactive and non-intrusive, allowing users to decide when to request support and how to respond to AI output.	Human/AI-in-the-loop interaction; user control; reflective workflow; non-intrusive AI support.	Frontend Interaction/UI Layer + Workflow Logic
Cycle 5: Formative Technical Evaluation	Reviewed prototype use through Technical Validation. Used observations only to guide future refinement.	Areas for improvement, including usability friction, prompting flow, AI-output interpretation, and workflow timing. These were not treated as formal research findings.	Prototype refinement; usability review; future iteration planning.	Evaluation / Prototype Refinement Layer

Table a: Showcases the different cycles as well as their impact on the application and overall architecture of the program

Design Decision around Generative AI Usage

The decision to utilize a specific application of multimodal Generative AI within a system workflow (rather than standard text-to-image or sketch-to-image) was based on the cognitive limitations of single generative interactions. Purely text based AI models rely on the designer's ability to convey language and then rely on the models ability to recognize the language and to produce manipulations of visual elements that may limit

creative freedom [2]. And on the other hand, using just sketch-to-image tools often lack the specific conventions or terminology that a designer might need to rely on when conveying higher cognitive parameters related to their design [2].

This impacted the design decision of incorporating a multimodal approach to the workflow system that would need to accommodate naturalistic ideation. This meant that the user would need to rely on sketching when their ideas are abstract and to have the ability to incorporate text prompts to refine their ideas more concretely. This implementation was intentional as it allows the human to have more decision-making power while also limiting their cognitive load as the AI assistant would make sure to only be able to assist the creative process without overtaking it [2]. Finally the AI is implemented to explicitly produce an abstract “sketch-like” image as it provides an inspirational stimulus rather than an overly realistic image that can limit the user’s creative freedom through design fixation [18].

System Architecture and Workflow Diagram

To facilitate a collaborative relationship between the Generative AI tool and the user the workflow was structured based on a “See-Transform-See” Loop [2]. For this project the process the application incorporates is instead a “*Sketch-Convey-Reflect*” Loop. This differs because the user can sketch their rough ideas, they can then convey their concepts with further descriptive text prompts that will provide them an output that they can reflect upon. This ensures that the AI acts only as a reactive partner that responds to the users inputs declaratively without disrupting their creative flow [2].

The Step-by-Step Multimodal Feedback Loop:

- **Step 1: User Externalization (Sketch/Text Prompts):** The user begins to conceptualize their ideas onto the digital canvas however they are also at liberty to express their ideas through textual prompts to provide further descriptions. Their prompts are then provided as inputs for visual data (PNG) and semantic data (Text).
- **Step 2: AI Interpretation (Convey):** The system takes in both prompts and simultaneously processes the sketch and text. The generative model analyzes the spatial attributes of the sketch as well as the contextual text prompts.
- **Step 3: System Generation (Convey):** The AI then generated an abstract suggestive image on a layer beneath the user's working layer. The image is based on the combined inputs from the previous step and it is applied with constraints that encourage the image to be lower fidelity and to resemble sketching properties.
- **Step 4: AI Communication (Reflect):** The image is provided on the reference layer and is automatically applied with an opacity attribute so as to clearly differentiate from the user's working layer. If there is already a previous AI Image the system will change the older generative image visibility off so that only 1 reference image is displayed underneath

- **Step 5: User Reflection (Sketch):** The user can then engage in a reflective conversation with the output of the AI system. They can evaluate the AI's interpretation of their concept and manipulate only the position and size of the AI image. This supports the user to only refer to the image to help them “scaffold” or deviate from it if necessary.

Design Requirements

Development of the user experiences started with a set of design requirements based on the literature review, which influenced the approach to the frontend and backend development. The central requirements include:

1. Accessibility and Low Barrier to Entry [27]
2. Assistive Technology Approach
3. Multimodal Feedback Loop (“*Human-in-the-loop*”)
4. Layered Mental Model Representation

Accessibility and Hardware Independence

The critical component of this project is based around the intrusiveness of technology's impact on the user and how to manage/minimize it. This tool is meant to be most effective when it achieves a “readiness-to-hand” which is when the user's interactions are transparent and they are allowed to focus entirely on the task instead of the mechanisms of the tools [21]. This led to the use case of supporting the digital framework around web interfaces due to multiple pragmatic factors. To name a few this approach meant that there would be a lower barrier to entry to engage with the tool as it would no longer be as hardware dependent as it would take advantage of any microprocessor as long as there would be a web browser interface.

Assistive Technological Approach

Also due to how this project revolves around incorporating a large language model as well as multimodal input and providing generative Ai as an output, meant it would have relatively high computational expenses. This meant that utilizing current market cloud services (Google Gemini API in this case) would increase the accessibility of the tool to more designers. And it just so happens that this approach is similar to Distributed Cognition. While utilizing the application the designer can offload their cognitive load onto the AI to assist them with conceptualizing their internal mental model but on the technical side the Application within the browser distributes the computation to cloud services. The user can then engage with the program while the complex generative tasks can be returned back to them, thus encouraging iterative sketching [32].

Multimodal Feedback Loop

The requirement for implementing an Ai assistive tool that enables the user to complete and finish sketching their mental model is based on the theory of fluid concepts [7]. That means that when the user receives an AI-generated reference model, they are subject to experience “conceptual slippage” where the designer will refer to the generative layer while they sketch. And because there is a separation between the AI layer and the user's working layer the design workflow encourages the back and forth interaction the designer will have

when using AI to support their designs.

This feedback loop is impactful for the design process as human drawing is for the most part sequential and dynamic since the individual can only draw iteratively with one stroke at a time all the while incorporating visual feedback [32]. This workflow encourages the user to engage in a continuous dialogue with the system in the following ways: they sketch their idea, they then describe it and the system interprets it and communicates it back to the user. This acts as a “reflective conversation” between the user and their tools [27]. Because this process makes use of a multimodal input (text and image) the AI can act as an inspirational tool to prompt the designer to recognize new creative avenues or possibilities.

Layered Representations of Mental Models

for this AI tool to support the designers internal mental model, it has to be non-intrusive and embrace the layered workspace. Mental models are dynamic structures that are created on the spot, almost adhoc and they rely heavily on the individual's underlying conceptual background understanding (12, 15) . This program makes use of a “Layered Transparency” as there is a three layered canvas stack for the user to work with (User Sketch, Generative AI Layer, Background Canvas Workspace) as a way to separate the Users attempt at expressing their internal mental model from that of the AI’s interpretation of it.

This layered architecture is an intentional design decision that is integral to this project. That is because the user is still at liberty to interpret the AI layer but because they cannot edit or export it, they can still work on refining their sketch since it will be the output. Thus if the user is referring to the AI underlay they are essentially treating it as a sketching scaffolding prompt to refine their initial sketch. Previous studies have revealed that converting high-fidelity AI designs into lower-fidelity provides a sketch scaffold when introduced through a transparent underlay [18]. This functions much like a digital version of tracing paper and non-intrusively guides the user without directly overriding their control over the final outcome. This approach also benefits as it mitigates “AI Fixation” as the designer is no longer constrained by the AI’s output if it were to appear too complete [18].

MSM System Architecture

This theoretical and thematic guidance led to four architectural elements for this project:

1. **Core Engine Foundation:** Focused on maximizing accessibility and achieving "readiness-to-hand" interaction by utilizing HTML5 Canvas API for a web-based, hardware-independent interface. This architecture supports zero install deployment and can push updates to ensure designers can continue sketches with AI image generations happening in the background
2. **Multimodal Integration Approach:** Incorporated a multimodal feedback loop to differentiate the system by accepting both visual (user sketch) and textual inputs, utilizing the

gemini-2.5-flash-image-preview model for its generative and multimodal capabilities. This flexibility means that the user can rely more or less on visual or semantic skills.

3. **Multi-Layered Architecture:** Implemented a **Layered Transparency** approach using a three-layered canvas stack (User Sketch, Generative AI Layer, Background Canvas). This intentional design decision separates the user's editable work from the AI's reference model to preserve the designer's authorship and mitigate **Design Fixation**. By acting as a turn-based partner it lowers the chances of cognitive overload on the user.
4. **Interaction Refinement and Feedback System:** The design workflow was finalized as an **AI-in-the-Loop** model to encourage a continuous "reflective conversation" with the material, enabling the user to intentionally deviate from or trace the AI-generated "sketch scaffold". The interface lends itself to support the dynamic dialogue with the AI agent working only when prompted by the user.

Frontend

The system is designed to support users in developing sketches that more closely reflect their internal mental models. For this reason, MSM emphasizes the relationship between the user's intended concept, their sketch, and the AI-assisted reference output. When the generated reference model more closely aligns with the user's intended mental model, it may provide a clearer scaffold for refining the sketch.

The interface is informed by the Strange Loop model [7], which describes a feedback loop between the user and the system. In MSM, this loop occurs across three connected representations: the user's sketch, the AI-generated reference model, and the user's internal mental model [26]. The designer moves between these representations by sketching, prompting, interpreting the AI output, and revising their work.

This interaction is supported through three system layers. First, the input layer allows the user to provide both a sketch and a text description of their idea. Second, the processing layer sends the visual and textual data to the AI model for interpretation. Third, the output layer presents a low-opacity AI-generated reference image beneath the user's working layer. This underlay is intended to support reflection and refinement without replacing the user's own sketching process.

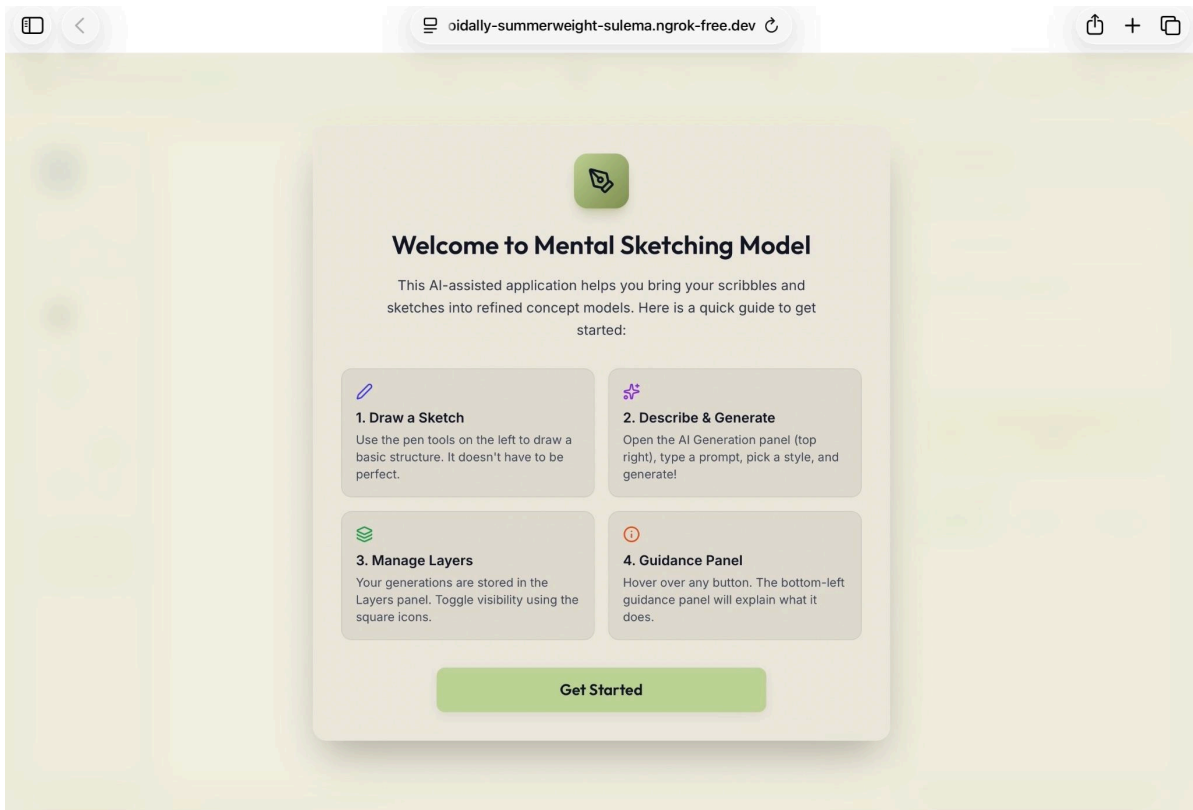


Figure A: Opening Welcome state of the Application

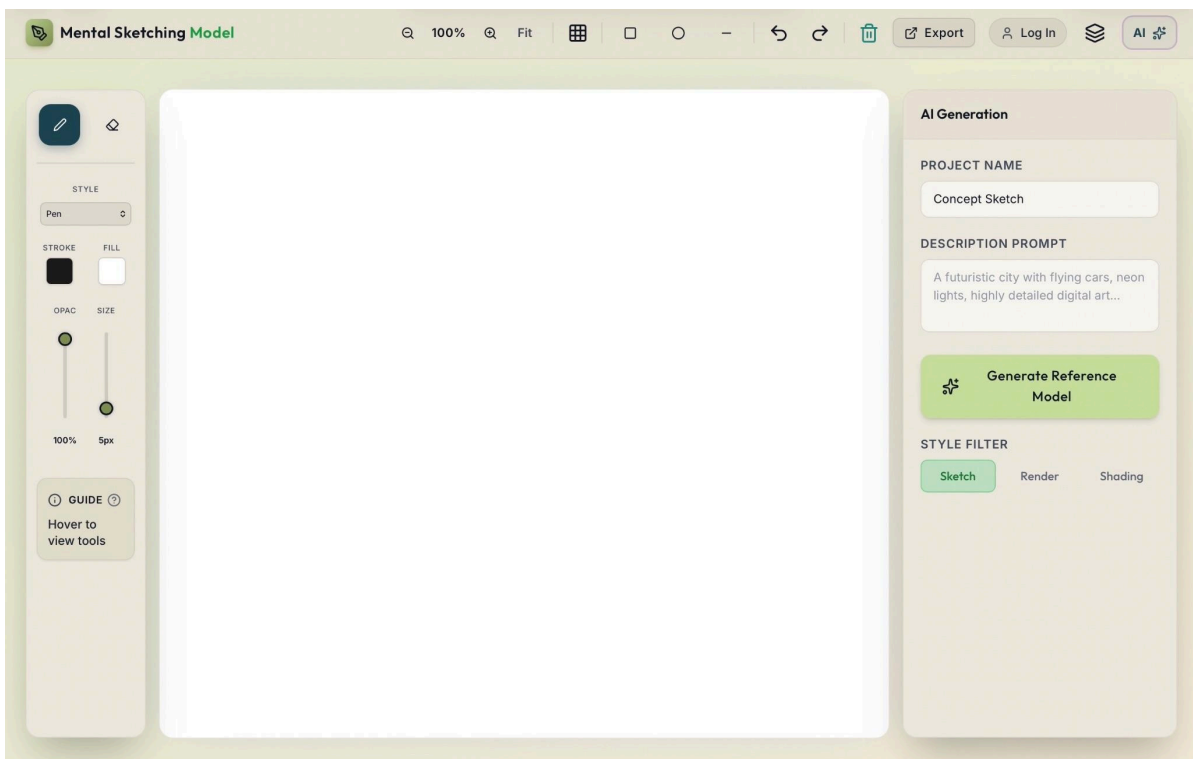


Figure B: the main page for the application with the tools on the right, canvas in the middle and AI Panel on

the right (can toggle between AI and Layers)

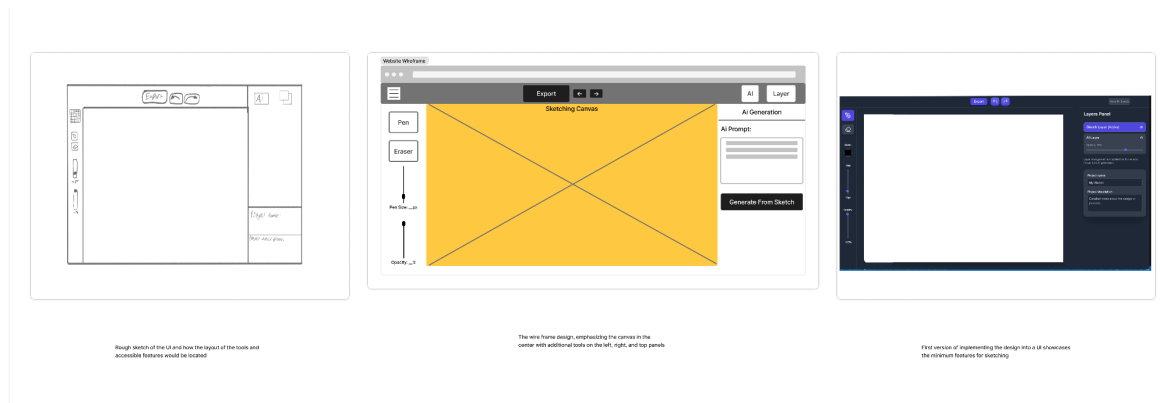


Figure C: (Frame 1: Rough sketch of the UI and how the layout of the tools and accessible features would be located) (Frame 2: The wire frame design, emphasizing the canvas in the center with additional tools on the left, right, and top panels)(Frame 3: First version of implementing the design into a UI showcases the minimum features for sketching)

Proof of Concept: Technical Validation Walkthrough

Phase 0: System initialization and Environment

The system utilizes tunneling services in order to connect to the web components for the application, and so there is an additional preliminary step that is required in order to set up and run the prototype for this application. In light of a hosted tool using a provided server system, the tunneling service “ngrok” was used as an affordable and rapid development implementation to share and distribute access to the application based on HTML distribution. Within the Google AntiGravity Coding IDE environment both known Node.JS and Express have to be running through the terminal in order to connect the services to the developer computer In order to gain access to the application from any other device with the provider link.

Upon getting access to the lake, the very first introduction to the application includes a welcome panel that the user has to acknowledge before continuing to the main page. The main page consists of a blank canvas at the center with a toolbar on the left side and a layers panel on the right. Additionally, there are assistive features on the top header that consist of shapes as well as controls for undo, redo and export. There is an additional section within the bottom left of the tools panel that is to help guide users. This automatically updates based on where the cursor or user is hovering over as a way to inform the user on the feature they are about to select. Once visiting the link, the only requested task upon the user in order to continue proceeding would be clicking the “get started” button in order to proceed with sketching. Because that is the only required action this confirms that the system successfully achieves a zero installed deployment that essentially removes any common friction such as downloading specific software. See figure a, and figure b.

Phase 1: Externalization (Sketch+Text Input)

The application consists of a blank canvas there is nothing for the AI to process. This means that there needs to be some sort of spatial input in the form of providing a sketch. This sketch needs can be displayed on the canvas as the user draws and additional modifications can take place on top of the drawing. Such modifications extend from switching between the pen and erasing tool as well as adjusting their color and size. Within the top header of the window application. There is also control for history with undo and redo buttons. The minimum requirement to run the generative AI layer is a tax prompt, however, for the sake of Conveying the mental moral Both a sketch and a tax prompt should be included in order to provide enough detail of the concept.

This phase relies on the capability of the AI system to externalize the designers internal mental model. As a proof of concept, the construction of a cylinder was situated within the canvas in a semi perspective matter. This was rendered through HTML5 Canvas (see figure e). The reason for this object to be drawn in such a manner was to demonstrate how the user can provide an imperfect concept of a cylinder where the lines are not smooth, nor is the front in the back end of the Cylinder indicate that the design designer has good concepts of perspective, sketching. To assist the AI to be able to understand the intent of the designer goals the text “perspective” was included within the text box. This is important as it establishes the context of the objective the designer has in mind with what they have drawn and in this case, their desire to refine their perspective, sketching skill. This phase demonstrates how the MSM can successfully bridge a gap between the users, low fidelity sketching with structured text data.

Once the generate button is clicked the following happens within the background of the application:

1. The application no longer becomes interactive as a loading icon is displayed thus indicating that background processing is occurring and to discourage the user from making any adjustments or changes that might disrupt the process.
2. The sketch layer is sterilized into a base 64 PNG string. This is to make sure that the specific line weights and hand drawn qualities of the sketch is maintained in order to keep the designer styles and perceived as intended.
3. The system then combines the base64 string alongside the text string in order to produce a JSON payload.
4. This is then transmitted via a post request to the NODE.JS proxy server. The step is important as the Google cloud API is responsible for preparing the data for the Gemini-Imagen pipeline.

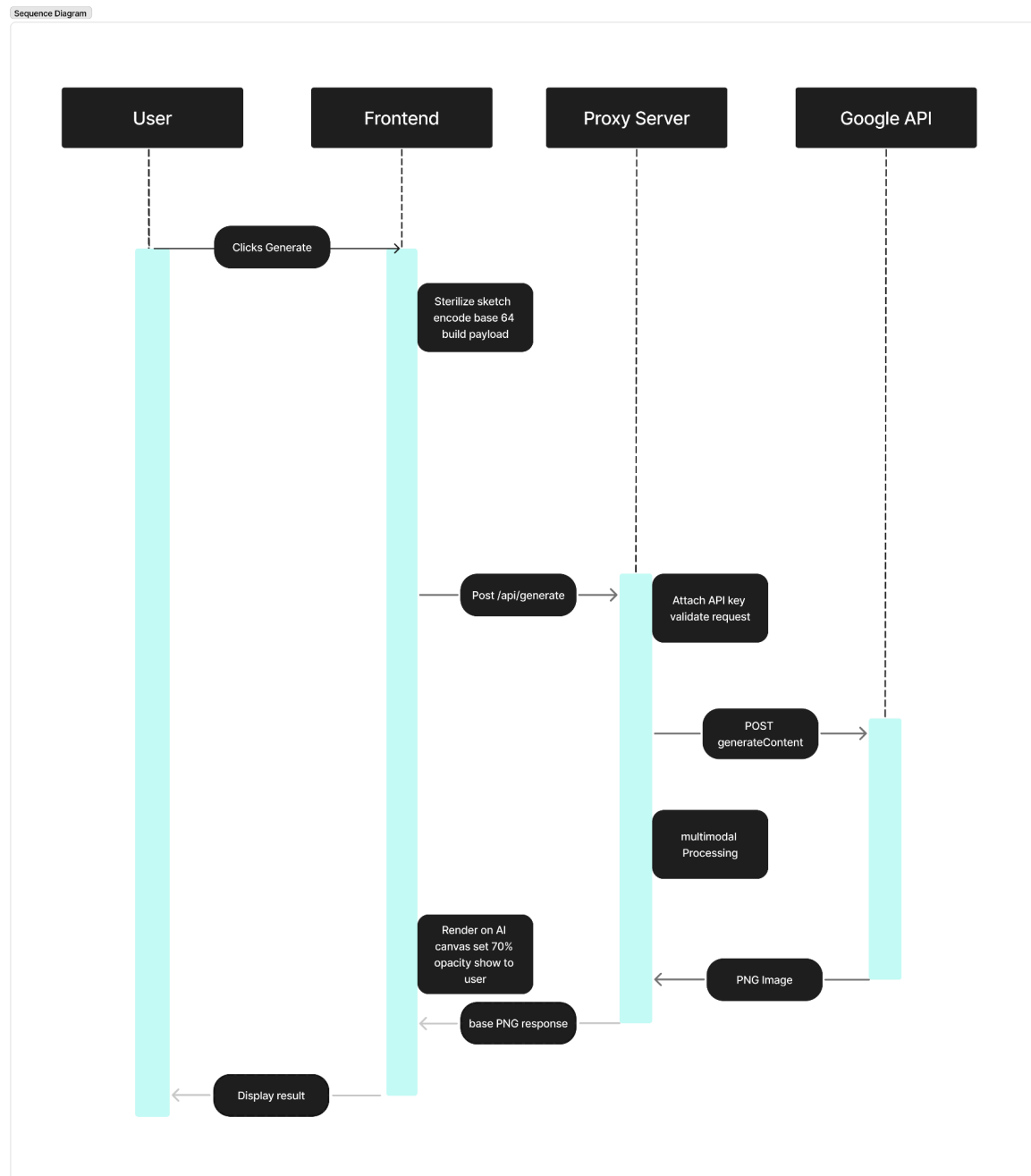


Figure D: Here is the sequence diagram for indicating how information is processed from the user to the front end and servers all the way to the API and back again to the user.

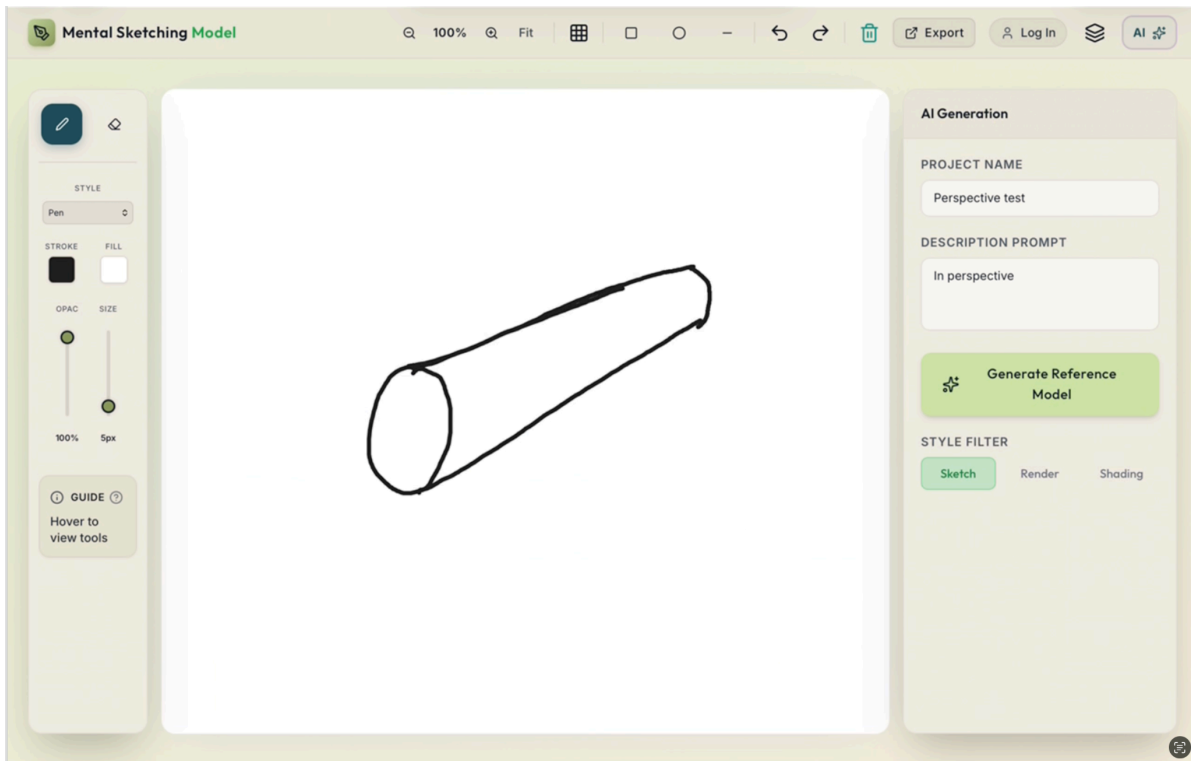


Figure E: This image contains a cylinder in perspective. Without the additional text prompt, there's less grounding to interpret from this as it could be misinterpreted as either a log or an axle or anything else.

Phase 2: Chained AI Logic (The Two-Phase Pipeline)

This phase is relevant to how the system validates its grounding capabilities. It goes beyond simply image generation and can demonstrate how the system interprets hand-drawn geometry as a structural constraint for the AI. Once the Multimodal payload is received on the backend, The system makes use of Google Gemini 2.5 flash as a way to perform a vision based analysis of the sketch. As opposed to other image to image models, this implementation does not overwrite the user input. With regards to this example, the model is tasked with analyzing orientation, assuming that points and elliptical curves are relevant to the cylinder. Testing this walk-through confirms that Gemini effectively translated the sketches. If there were any errors, it would've been reported that the sketch or the text were not able to be processed through the AI service. This would also result in the application providing an Error message if there was any possibility that the Service was not able to connect. See figure d.

Phase 3: Suggested Scaffolding (The Output)

This phase focuses on the synthesis and delivery of the AI generated scaffold and validates that the Output be provided as reflective material that does not impede on the final outcome of the sketching process. The prompt that was generated in a previous Phase is transmitted to "Google Imagen 4" in order to be synthesized. The resulting image is then of course returned back to the client side interface as a B64 string.

However, instead of immediately displaying the image, there is a confirmation window that pops up prior to being able to view the output. This provides a reflective pause that the user needs to take into consideration for future image generations. This ensures that the user remains as an active agent as they are engaged with being able to choose exactly when to reflect on the AI suggestions.

The validation of this phase is indicated through the output of the AI generation of a low Fidelity abstracted reference rather than a photo realistic render. This abstraction is necessary in order to relate the image generation more closely to the user sketch. Through the sketch-like underlay the system avoids design fixation where the user might be compelled to just copy a perfect AI rendered image. This follows in line with providing support without explicitly indicating what features the design should have on its final outcome. See figure f. Also see figure h.

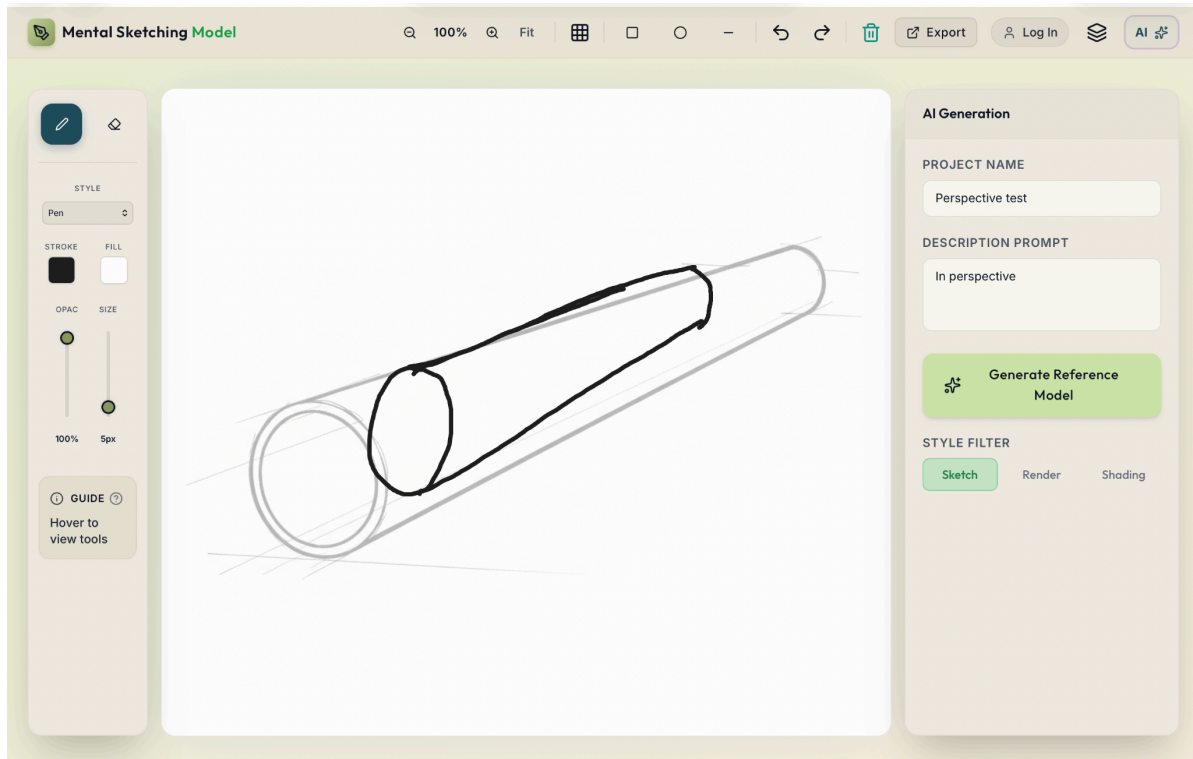


Figure f: This reference shows how the AI is able to interpret That the image is a cylinder, however, because there was not enough information describing whether it was hollow or not, the AI took liberty of expanding upon a hollowed reference model.

Phase 4: Reflective Interaction (User Agency and Authorship)

The spinal phase of the walk-through validates the multilayered architecture in its ability to support a non-intrusive human author design. The generated reference image is injected into an “AI layer” which sits beneath the “user sketch layer” however it is above the background canvas. The interface automatically provides an opacity of 30% in addition to including a transparency slider within the layers panel. This is to

enable and allow the user to adjust the opacity of that layer in order to provide an ideal tracing paper experience. When the opacity is lower, there's less incentive to sketch as well as less confusion over AI generated lines and user sketch lines.

Throughout the walk-through, the AI provides a cylinder as a scaffold or reference that can help refine the ellipses and orientation of the “user layer” sketch. Because the layers are isolated, the user can choose to either incorporate attributes from the AI suggestion or to deviate from it. Of course, all this can be regenerated as a method to provide a new interpretation of the user sketch and text prompt. This phase is validated since the isolation of layers proves that the system addresses a separation of the user and AI without impeding on the user sketching layer. This is further enforced as the final design remains authentic to the capabilities of the designers sketching skills. This is because the export button at the top of the header will only export the user layer. This ensures that the final outcome retains human authorship while also containing an AI loop as the design is capable of receiving AI input, if explicitly acted upon by the designer. See figure g.

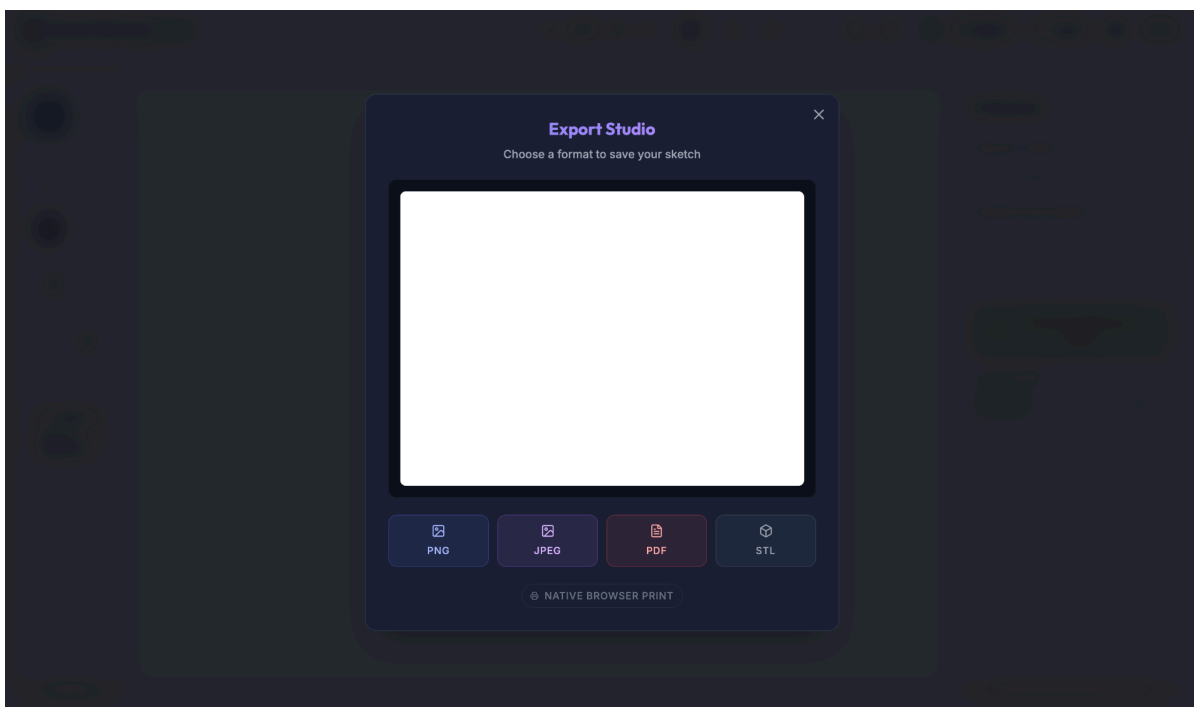


Figure g: This is the expert Studio, which allows for different formats for exporting the image. This is left intentionally blank to showcase how the ui and layout of the export capability.

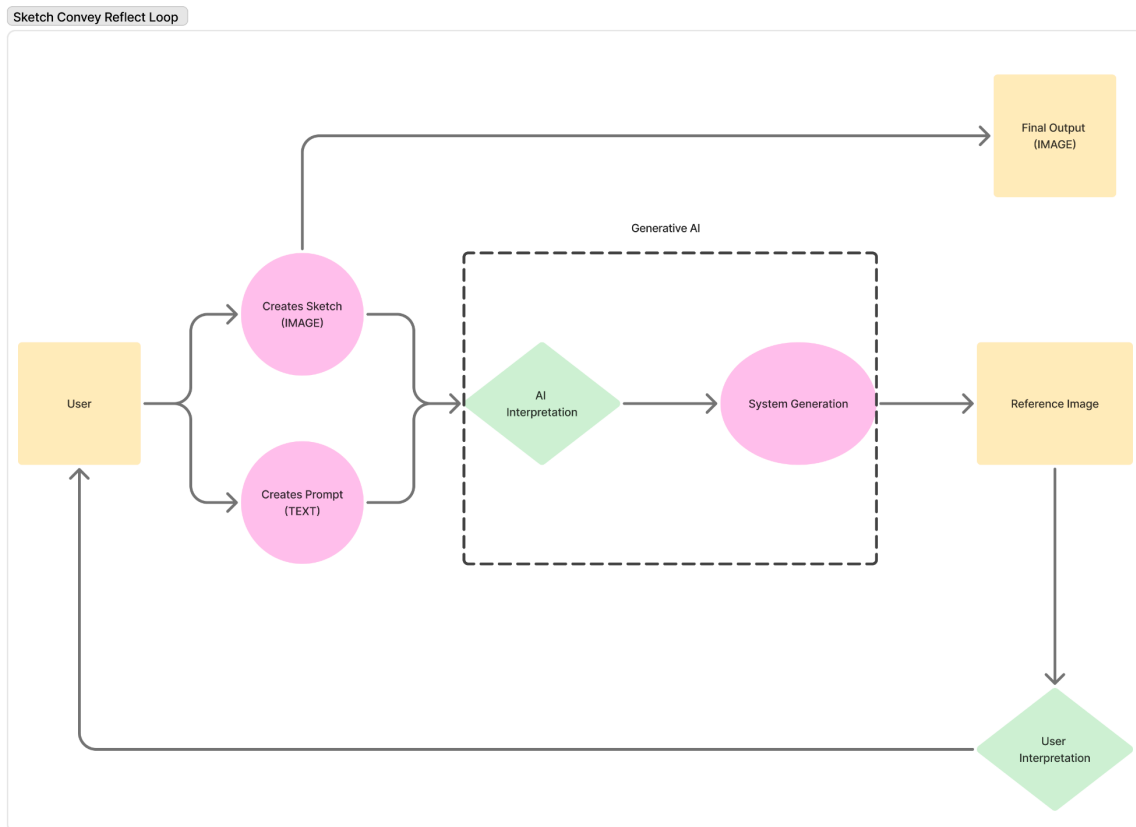


Figure H: This is the sketch conveyor reflect loop that indicates how the user provides information through generative AI and uses it as a reference for creating their own final output.

Backend

Generative Ai Model and API

To reiterate on this project's scope: this is an application that integrates modern Web Technologies with advanced **Generative AI Services** to create an assistive user experience. **Generative AI** used in this paper refers to artificial intelligence capable of creating new content (like images or text) rather than just classifying or analyzing existing data. The coding was done through Microsoft Visual Studio Code, which is due to the fact that it can enable the use of AI assisted editors for any coding challenges. This is primarily used for connecting to APIs and enabling Ai models. The program utilizes “gemini-2.5-flash-image-preview” which is classified as a Generative Multimodal Model. (hence why it's taking both visual and text input). The program utilizes “Google Generative Language API” for textual input and the program uses HTML's “Canvas” capabilities that can allow for sketching within web pages. Multimodal means that the AI can understand and process more than one type of data input simultaneously (examples would be text, images, audio, etc...). And in the case of this project, it handles both **Visual Input** (the user's sketch) and **Textual Input** (the detailed description). Now because this project makes use of the models that are designed to accept both text and image inputs it

incorporates a process known as “grounding” or “Multimodal Prompting” to generate the new reference image. The term grounding is when the AI is instructed to base its output on a specific piece of input data (the designer’s sketch in this application) rather than creating an image entirely from scratch. Multimodal Prompting is the combination that allows for precise creative control over the final output of the AI. Such as when using the sketch for the structure it can build upon the use of text prompts to more accurately depict what the User has in mind.

API Endpoint

The API that this project connects to is the “Google Generative Language API”. Just to clarify the use of this API is the acronym stands for: Application Programming Interface. This means that it will take the request from the Designers actions and bring a response back to them without needing to understand all of the intricacies of how the AI model behaves. This utilized an api endpoint that holds a specific address (URL) that is within the server for where the application will then send its request for a task (in this case generating content). This method is done through a “POST” which is often used whenever the application has to send a significant amount of data (for example both image and text prompt) to a server to create new data.

Code Distribution

The project source code would normally be distributed through GitHub to showcase its development lifecycle. This would include but not limited to the initiative, comments and the evolution of the software as it was being developed. However, due to the limited time constraint and the approach for rapid prototyping, the API key was hard coded into the project files to facilitate rapid implementation and testing. This means that the approach presented security risks if it were to be hosted on GitHub, whether if it was private or public. That means that the project files are currently stored locally on a personal computer and available for display upon request.

Discussion

This application project: Mental Sketching Model (MSM), serves as a multimodal approach as a non-intrusive AI assistive tool designed to support designers during the early stages of ideation. This project implements a two-phase AI pipeline that uses Gemini 2.5 Flash for spatial grounding and Imagen 4 for low-fidelity synthesis. This system acts as a reflective creative mirror rather than an automated AI image generator. The architecture in this application prioritizes a layered non-intrusive workflow where the AI output remains a separate (scaffolded) layer so that the designer retains their authorship and agency throughout the ideation process. The iterative design process provided insights into the three core research questions. In terms of sketching and learning (RQ1), this project demonstrated that the designer's intention is best captured through a combination of spatial sketching and semantic text. This multimodal approach allowed the system to interpret messy, low-fidelity input and produce a reference model image that accurately grounded the AI output within the

designer's goals. With regard to assistive technologies and cognitive impact (RQ2), this project implemented a layered architecture that was critical in limiting design fixation. By providing an abstracted sketch underlay rather than a high-fidelity render, this system encourages the designer to use the AI as a reference to be traced (or of course deviated from) rather than being copied. The deliberate friction introduced via the confirmation window served as a cognitive guardrail. This is because it prevents rapid unreflective AI prompting. Finally with respect to AI as a pedagogical tool (RQ3), the position of AI as an active partner in the loop successfully supported the iterative cycle of externalization. The system did not dictate the creative direction. It instead provided a situational stimulus that helped designers move past the blank canvas barrier while maintaining a reflective conversation with their material.

A significant unexpected outcome was the profound impact of deliberate friction. Initially, the confirmation window was a minor interface choice, but it emerged as a mechanism discouraging excessive AI generations. This intentional pause proved as a way to encourage reflection on the sketching process, allowing the designer to compare their internal mental model with the AI's externalized interpretation. Conversely, early iterations revealed that high-fidelity AI output actually hindered the early design process. High-fidelity images introduced visual noise and features that were not explicitly indicated. Shifting the system focus toward low-fidelity, stylistically matched sketches aligns this project as a pedagogical support tool that speaks the same visual language as the novice designer.

While this project confirms the value of non-intrusive AI as seen in related scaffolding literature, the findings are limited by the exploratory nature of the Research through Design methodology. A technical validation uses a single-designer behind the software to indicate and define the logic of the system but does not yet provide empirical evidence of its effectiveness across diverse user groups. Future work should involve a larger scale mixed-methods study with approximately 50 participants to move from exploratory reflection toward empirical validation. This study would shift from defining interaction principles to evaluating its capabilities across individuals with varying skill levels. Interviews and analysis could examine the relationship between user's intentionality and perceived AI alignment. This could help with specifically focusing on how well the AI respects the construction lines and perspective of the user. The system's data collection could include quantitative tracking for user input and output and the relationships between the frequency of AI requests, and the timing of deliberate pauses.

Future iterations of MSM should introduce advanced pedagogical features such as automated construction lines. This could include construction lines and centerline axes or perspective grids. These could help students learn the underlying principles of form. This would require a specialized training set focused on construction drawing rather than finished products. Beyond just the technical requirements, there is a clear case for implementing MSM within design educational environments. Examples would be such as intensive workshops or studios for novice designers. A subscription-based service for design institutions could potentially provide a

sustainable financial model. This model could further benefit from the continued development of supportive guardrails that can enable human creativity, rather than replacing it. This project highlights a shift in Human-Computer Interaction: the goal of AI within design should not be just to automate but to empower users. By maintaining the designer at the center of the reflective conversation, this Mental Sketching Model (MSM) application supports the need to further develop the future of industrial design to be human-authored.

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